Habitat-GS: A High-Fidelity Navigation Simulator with Dynamic Gaussian Splatting

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Research Questions

- How can 3D Gaussian Splatting be integrated into embodied AI simulators to improve visual fidelity while maintaining real-time performance?
- Can photorealistic gaussian avatars serve as both visual entities and navigation obstacles to enable effective human-aware navigation training?
- Does training on higher-fidelity 3DGS scenes improve agent cross-domain generalization compared to traditional mesh-based environments?

Core Idea

Current embodied AI simulators use mesh-based rendering which lacks visual realism, creating a sim-to-real gap that hurts agent generalization. Additionally, existing human avatars are low-fidelity meshes that don't provide realistic training for human-aware navigation. This paper integrates 3D Gaussian Splatting into Habitat-Sim to achieve photorealistic rendering while maintaining real-time performance, and introduces gaussian avatars that simultaneously serve as high-fidelity visual entities and collision obstacles. The key insight is 'visual-navigation decoupling' - using 3DGS for photorealistic rendering while keeping traditional NavMesh for navigation logic, avoiding the complexity of physics simulation on gaussian representations.

Key Formula

Linear Blend Skinning for Gaussian Avatars

$$\mu'_i = \sum_j w_{i,j} \mathbf{T}_j \mu_i$$

Transforms canonical gaussian position μ_i to posed position μ'_i by blending joint transformations T_j weighted by skinning weights $w_{i,j}$. This enables real-time avatar deformation without neural network inference at runtime, crucial for maintaining simulation performance while achieving photorealistic avatar animation.

Key Figure

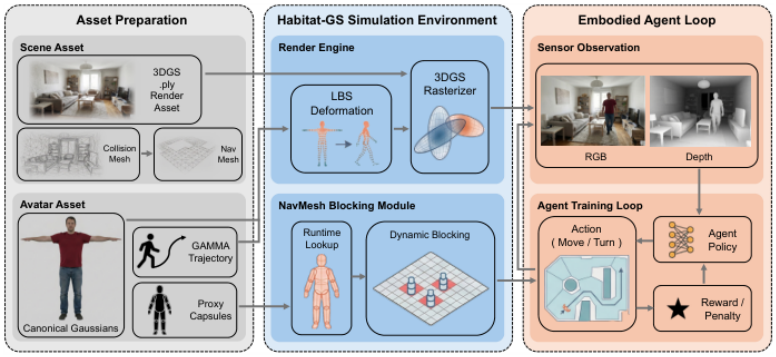

Fig. 2: System overview of Habitat-GS. From left to right: *Asset Preparation*, where 3DGS scene assets and gaussian avatar assets are prepared respectively. *Habitat-GS Simulation Environment*, where the render engine performs 3DGS rasterization for scene gaussians and LBS deformation followed by rasterization for avatar gaussians, producing RGB-D observations. The NavMesh blocking module retrieves pre-computed proxy capsules at runtime and injects them into the NavMesh for obstacle blocking. *Embodied Agent Loop*, where the agent policy consumes sensor observations, executes actions, and receives rewards shaped by both navigation success and avatar collision.

3 Habitat-GS Simulation Environment

Habitat-GS extends Habitat-Sim with two key upgrades: extending mesh-based scene rendering with 3D Gaussian Splatting (Sec. 3.2), and implementing a gaussian avatar module to drive and render avatars (Sec. 3.3). A central design principle underlying both modules is *visual–navigation decoupling*: 3DGS handles all visual rendering and traditional NavMeshes continue to govern navigation, which sidesteps the lack of explicit surface geometry inherent in the 3DGS representation. Below we discuss the overall system architecture and the detailed design of each module.

3.1 System Overview

Figure 2 illustrates the end-to-end data flow in three stages:

Figure 2: Shows the complete system architecture with visual-navigation decoupling principle, illustrating how 3DGS rendering and

gaussian avatars integrate with traditional NavMesh-based navigation.

Limitations

- Visual-navigation decoupling limits physical interaction to navigation-level obstacle avoidance rather than fine-grained manipulation tasks like grasping or pushing
- System requires pre-computation of proxy capsules and joint transformation matrices, adding offline processing overhead for avatar preparation
- Performance scales with gaussian count and avatar complexity, potentially limiting scalability for very large scenes or many simultaneous avatars
- Depends on CUDA hardware acceleration, making it less accessible on non-NVIDIA platforms compared to pure CPU-based mesh simulators